

Entropy: found

$$S = \frac{1}{T} \left[\frac{-k_B T V}{\pi^2} \int_0^\infty dk \, k^2 \ln(1 - e^{-\beta \hbar c k}) \right. \\ \left. + \frac{V \hbar c}{\pi^2} \int_0^\infty dk \, \frac{k^3 e^{-\beta \hbar c k}}{1 - e^{-\beta \hbar c k}} \right]$$

* Expand first integral

$$\int_0^\infty dk \, k^2 \ln(1 - e^{-\beta \hbar c k})$$

let $x = \beta \hbar c k$, then

$$\frac{dx}{\beta \hbar c} = dk$$

$$\rightarrow \int_0^\infty \left(\frac{dx}{\beta \hbar c} \right) \left(\frac{x}{\beta \hbar c} \right)^2 \ln(1 - e^{-x})$$

$$= \frac{1}{(\beta \hbar c)^3} \int_0^\infty dx \, x^2 \ln(1 - e^{-x})$$

: Integration by parts

* Product rule

$$\frac{d}{dx} (u(x)v(x)) = u'(x)v(x) + u(x)v'(x)$$

$$\text{where } u' = \frac{d}{dx} u$$

* integrate

$$\int \frac{d}{dx} (u v) dx = \int u' v dx + \int u v' dx$$

Thus

$$* \int u v' dx = [u v]_a^\infty - \int u' v dx \quad *$$

$$\text{let } u = \ln(1 - e^{-x})$$

$$u' = x^2$$

then

$$u = \frac{1}{3} x^3$$

$$u' = e^{-x} / (1 - e^{-x})$$

Plugging this into *

This goes to zero $\rightarrow 0$

$$\int x^2 \ln(1 - e^{-x}) dx = \left[\frac{1}{3} x^3 \ln(1 - e^{-x}) \right]_0^\infty$$

$$- \int \frac{1}{3} x^3 \frac{e^{-x}}{1 - e^{-x}} dx$$

Thus

$$\int_0^\infty k^2 \ln(1 - e^{-\beta \hbar c k}) = - \frac{1}{3(\beta \hbar c)^3} \int_0^\infty \frac{x^3 e^{-x}}{1 - e^{-x}} dx$$

Plugging this in:

$$S = \frac{1}{T} \left[\frac{k_B T V}{3 \pi^2 (\beta \hbar c)^3} \int_0^\infty \frac{x^3 e^{-x}}{1 - e^{-x}} dx \right.$$

$$\left. + \frac{V \hbar c}{\pi^2} \int_0^\infty dk \frac{k^3 e^{-\beta \hbar c k}}{1 - e^{-\beta \hbar c k}} dk \right]$$

Second integral:

as above: $x = \beta \hbar c k$

$$dk = \frac{dx}{(\beta \hbar c)}$$

$$\int_0^\infty \left(\frac{dx}{\beta \hbar c} \right) \left(\frac{x}{\beta \hbar c} \right)^3 \frac{e^{-x}}{1 - e^{-x}} dx$$

$$= \frac{1}{(\beta \hbar c)^4} \int_0^\infty \frac{x^3 e^{-x}}{1 - e^{-x}} dx$$

then

$$\int_0^\infty dk \frac{k^3 e^{-\beta \hbar c k}}{1 - e^{-\beta \hbar c k}} dk = \frac{1}{(\beta \hbar c)^4} \int_0^\infty \frac{x^3 e^{-x}}{1 - e^{-x}} dx$$

Plugging in:

$$S = \frac{1}{T} \left[\frac{k_B T V}{3 \pi^2 (\beta \hbar c)^3} + \frac{V \hbar c}{\pi^2 (\beta \hbar c)^4} \right] \int_0^\infty \frac{x^3 e^{-x}}{1 - e^{-x}} dx$$

factored out

$$S = \frac{V}{T (\beta \hbar c)^3 \pi^2} \left[\frac{k_B T}{3} + \frac{\hbar c}{\beta \hbar c} \right] \int_0^\infty \frac{x^3 e^{-x}}{1 - e^{-x}} dx$$

$$= \frac{V}{T (\beta \hbar c)^3 \pi^2} \left(\frac{1}{3\beta} + \frac{1}{\beta} \right) \int_0^\infty \frac{x^3 e^{-x}}{1 - e^{-x}} dx$$

$$S = \frac{4V}{3T \beta^4 (\hbar c)^3 \pi^2} \int_0^\infty \frac{x^3 e^{-x}}{1 - e^{-x}} dx$$



$$= \frac{\pi^4}{15}$$

$$S = \frac{4V \pi^2}{45 \beta^4 T (\hbar c)^3}$$

Correction to Notes: Pg 7-8

* on pg 7: Should be

$$U_{int} = + \frac{\partial}{\partial \beta} (BF)$$

* on pg 8: should be

$$S = \frac{1}{T} \left(- \frac{k_B T}{\pi^2} \int_0^\infty dk k^2 \ln(1 - e^{-\beta \hbar c k}) \right) + \frac{V \hbar c}{\pi^2} \int_0^\infty dk \frac{k^3 e^{-\beta \hbar c k}}{1 - e^{-\beta \hbar c k}}$$